

Möbius Polarization of Light

Ishir Dutta '17

Advisor: Enrique (Kiko) Galvez

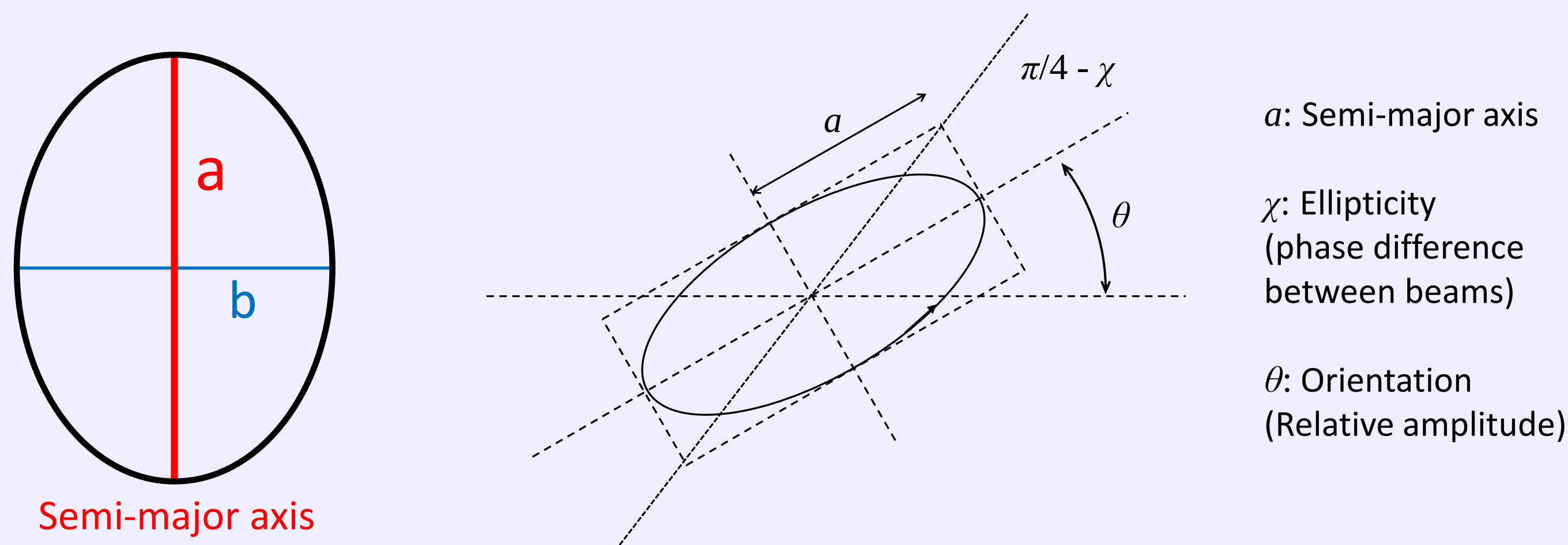
Department of Physics and Astronomy, Colgate University

Motivation

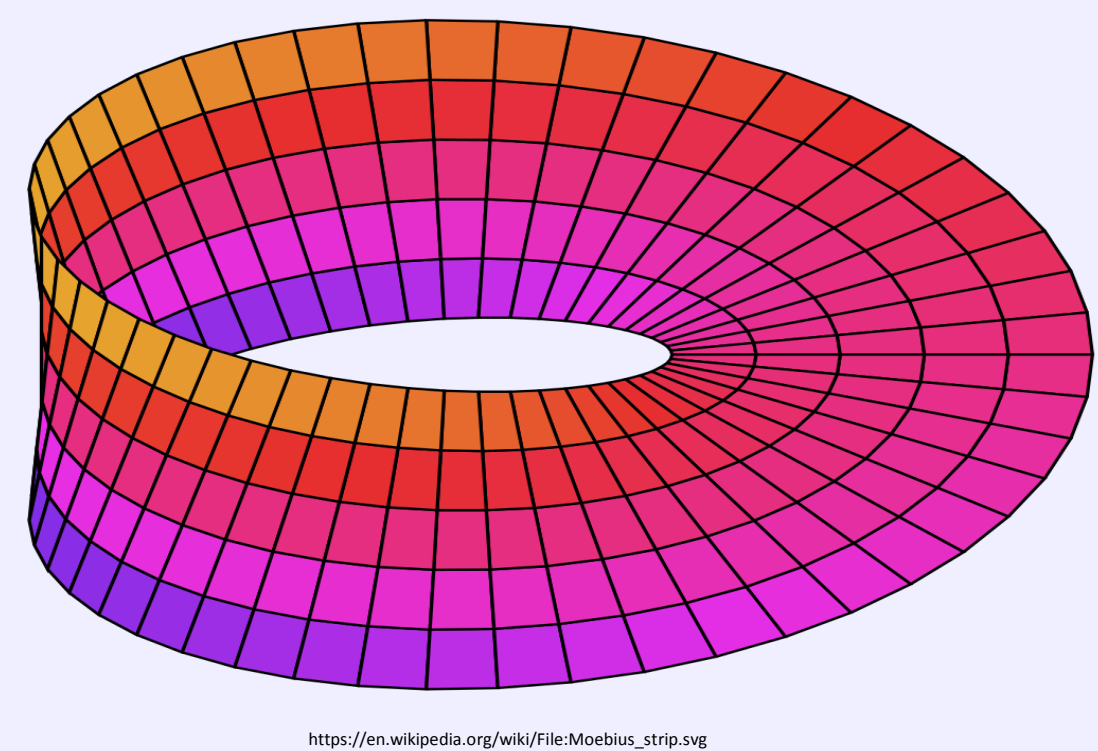
It has been theorized that interfering circularly polarized Laguerre-Gauss modes at small angles causes the semi-major axes of the resulting polarization ellipses to form Möbius strips.

Polarization Ellipses

- Polarization is the direction of the electric field of a wave
- May classify as linear, circular, or elliptical
- Elliptically polarized light is a combination of linear and circularly polarized light
- A polarization ellipse represents the shape traced by the electric field as it passes through a plane in space



Möbius Strip



$$x = \left(1 + \frac{v}{2} \cos \frac{u}{2}\right) \cos u$$

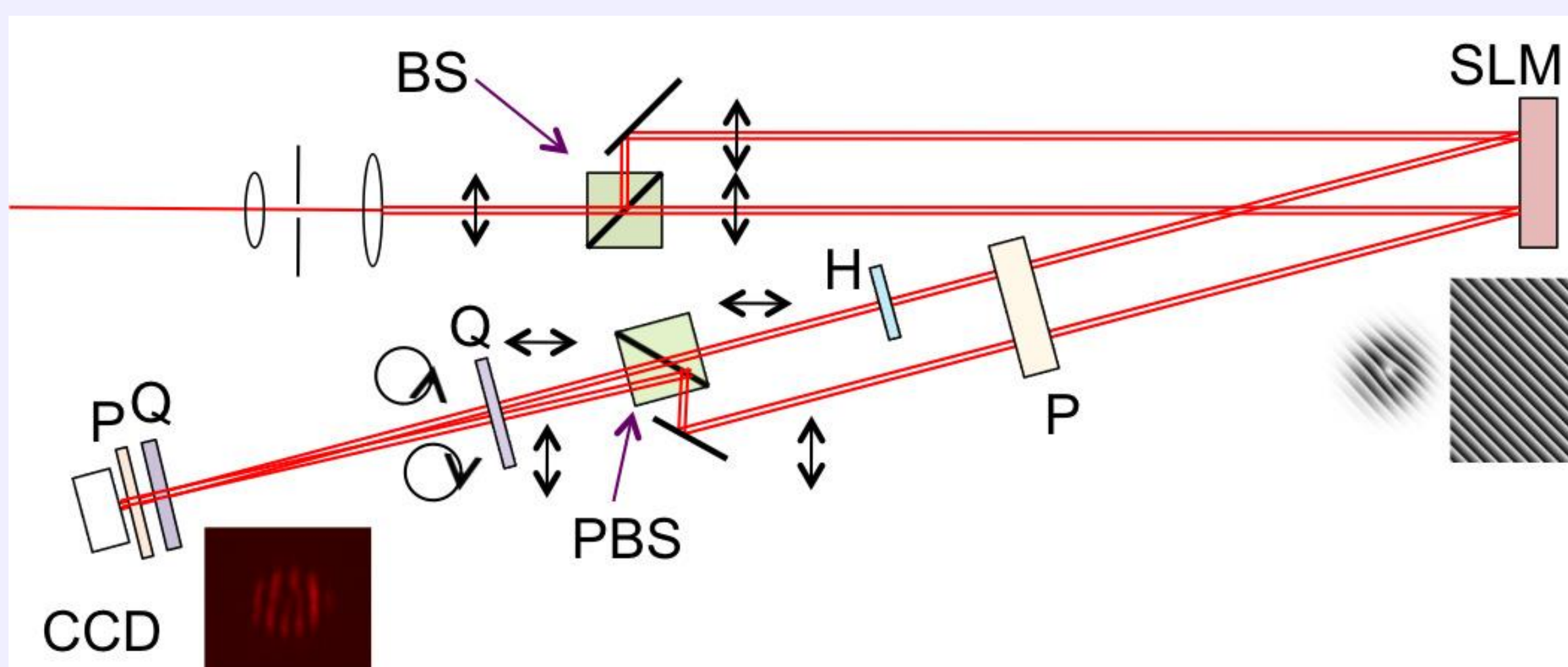
$$y = \left(1 + \frac{v}{2} \cos \frac{u}{2}\right) \sin u$$

$$z = \frac{v}{2} \sin \frac{u}{2}$$

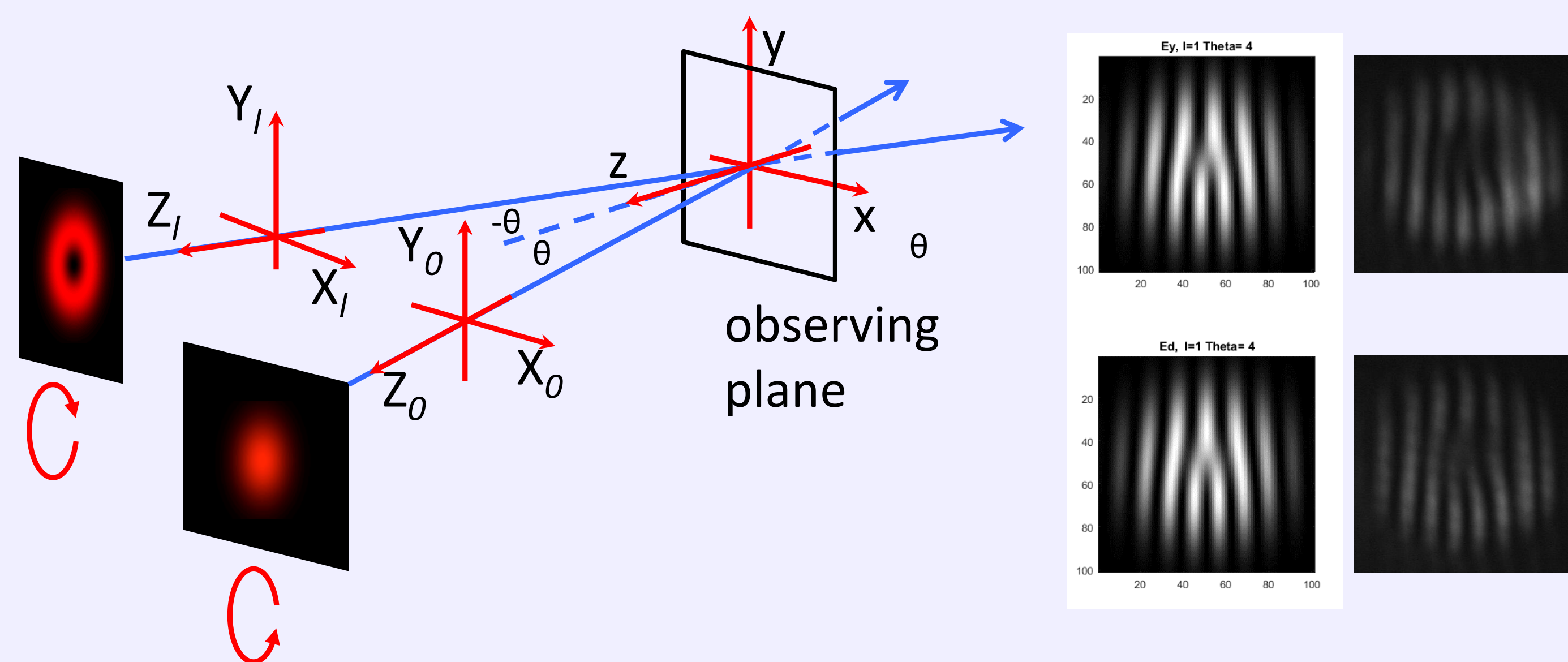
- A Möbius strip is a surface that has only one side and one edge
- A simple loop may be described by the given parametric equations
- Multi-twist loops can be created by joining two opposite edges of a rectangular surface while creating an odd number of half twists (Take one! →)

Past Research

- Research done previously at Colgate has found evidence for the existence of these Möbius loops using the following set up

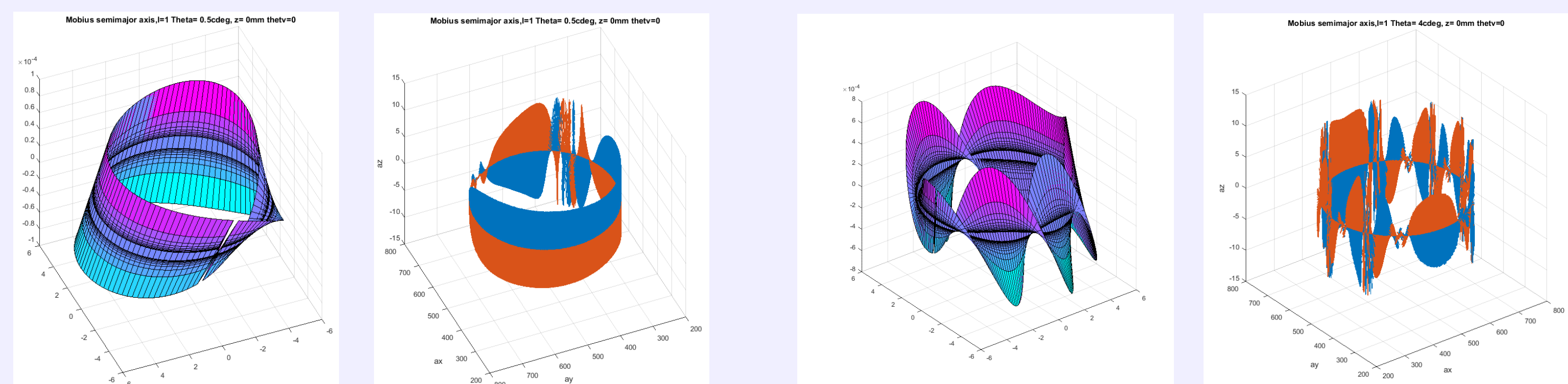


- Our analysis was based on the following coordinate system



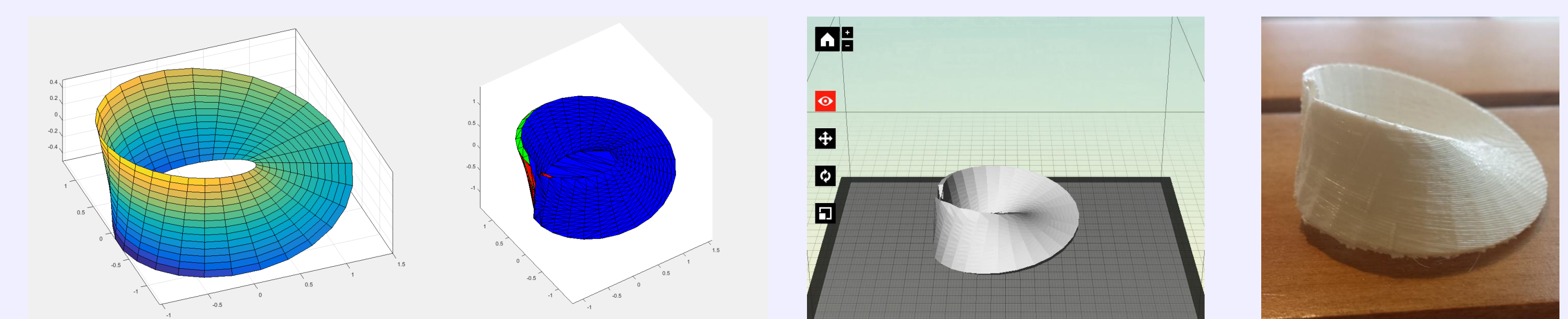
Data

- We used a MATLAB script to simulate the results of two interfering Laguerre-Gauss modes and visualize the semi-major axes of the polarization ellipses.
- We are working on another script, based on theoretical calculations of the amplitude and phase of the electric field, to analyse images of the two beams after passing through polarizers (H, V, D, A).
- We hope to use this code to reconstruct the Möbius strip from the fringe patterns shown previously.
- Some early results, showing the theoretical calculation (L) and the output of the analysis from the images (R) are below.

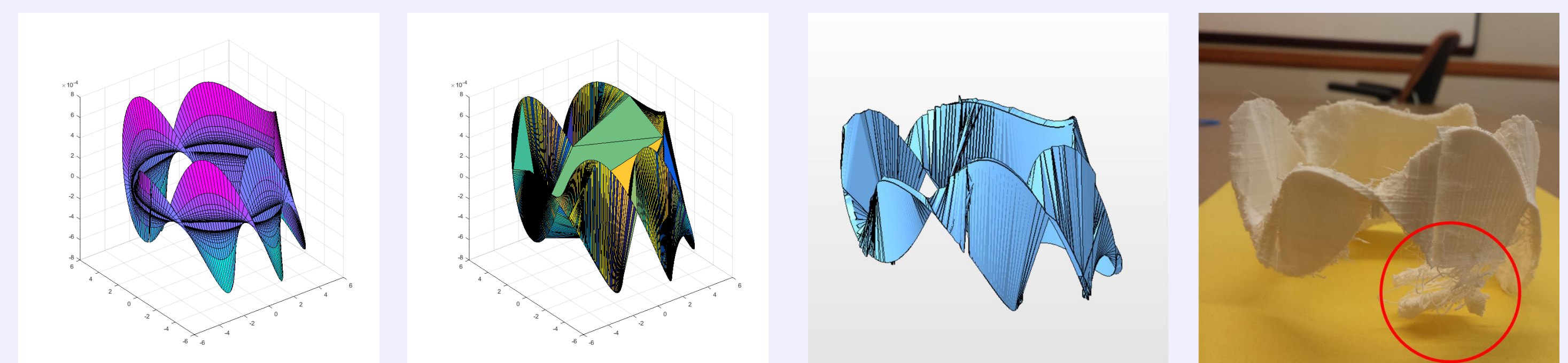


3D Printing

- 3D Printing is a method of manufacturing objects by slicing a computer model into layers and then forming those layers with a printer to generate the complete object.
- We used MATLAB functions 'surf2solid' and 'stlwrite' to give surface plots a thickness and to turn them into printable 3D objects, and Autodesk Netfabb to prepare the meshes for printing.
- We first printed a Möbius strip from the standard parametric equations: starting from the surface plot in MATLAB, to a 3D object tessellated by triangles, then a printable file, and finally a physical object (L-R).



- We then repeated this process for our simulated Möbius strips



References

- [1] Freund, I. "Multitwist optical Möbius strips," Opt. Lett. **35**, 148-150, (2010).
- [2] Galvez, E.J., Beach, K., Zeosky, J.J., Khajavi, B., "Space-variant polarization patterns of non-collinear Poincaré superpositions," Proceedings of SPIE, **9379**, 93790A 1-8, (2015).
- [3] Galvez, E.J., Khadka, S., Schubert, W.H., and Nomoto, S., "Poincaré-beam patterns produced by non-separable superpositions of Laguerre-Gauss and polarization modes of light," Appl. Opt., **51**, 2925-2934, (2012).
- [4] surf2solid from <https://www.mathworks.com/matlabcentral/fileexchange/42876-surf2solid-make-a-solid-volume-from-a-surface-for-3d-printing>
- [5] stlwrite is from <https://www.mathworks.com/matlabcentral/fileexchange/20922-stlwrite-filename--varargin->